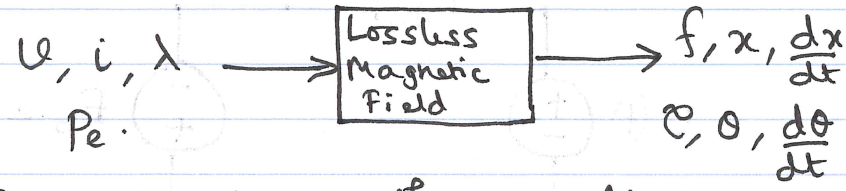
Lecture 17.

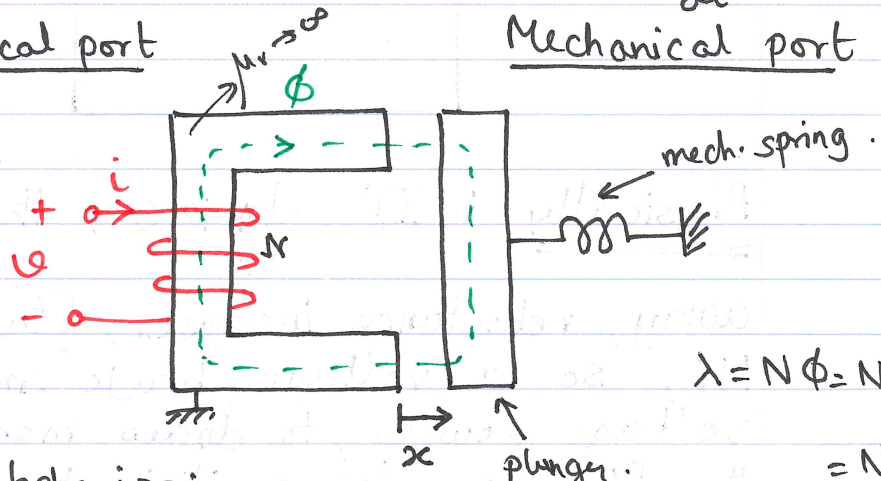
Last time : * In inductors, energy is stored predominantly in the air gap.

* Electromechanical energy conversion



Electrical port — Mechanical port.

*



$\lambda = N\Phi = N \cdot \frac{Ni}{R}$.

$= \frac{N^2}{R} i$

$= L(x)\, i$

Electrical port behavior:

$$v = \frac{d\lambda}{dt} = \frac{d}{dt} L(x) \cdot i$$

$$= i \cdot \frac{\partial L(x)}{\partial x} \cdot \frac{dx}{dt} + L(x) \frac{di}{dt}$$

$\frac{\partial L(x)}{\partial x} \cdot \frac{dx}{dt}$: speed voltage (velocity); $L(x) \frac{di}{dt}$: transformer voltage

In this example,

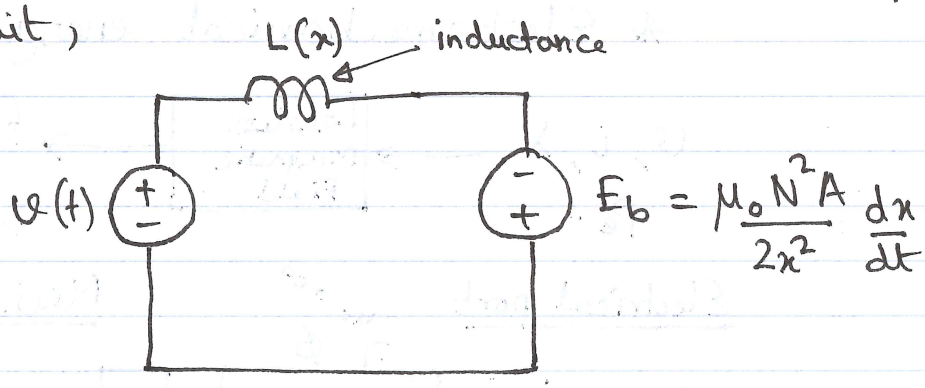
$$L(x) = \frac{N^2}{R(x)} = \frac{N^2 \mu_0 A}{2x}$$ [assuming $\mu_r \to \infty$]

super important.

transformer voltage | speed voltage

$$\therefore \quad v = \mu_0 \frac{N^2 A}{2x} \frac{di}{dt} - \mu_0 \frac{N^2 A}{2x^2} \cdot \frac{dx}{dt}$$

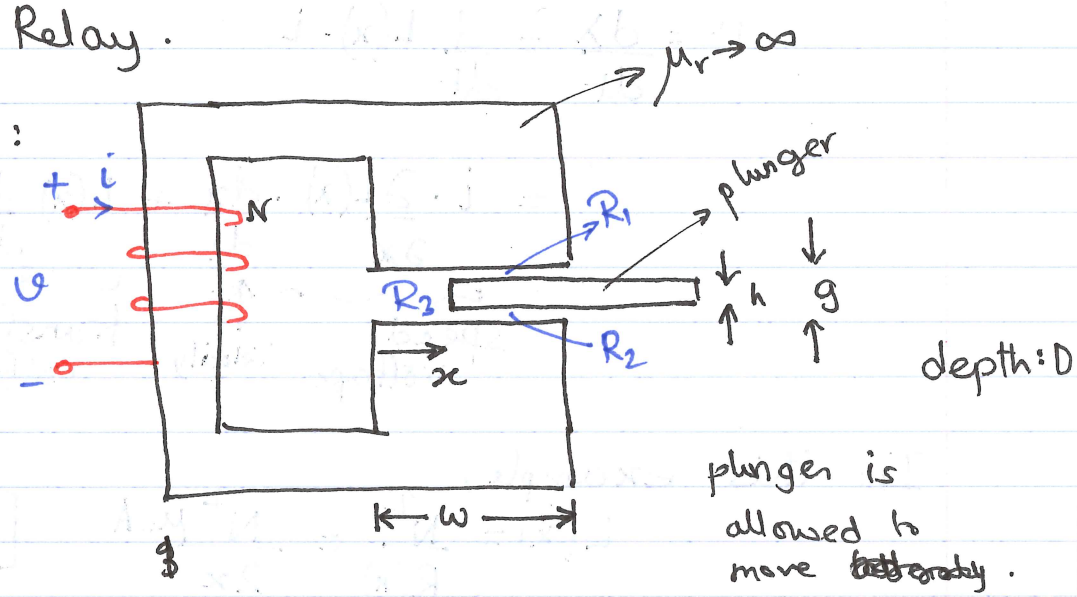
So, we can electrically draw an equivalent circuit,



Physically: If $\frac{dx}{dt} > 0$, the plunger is moving away, reluctance increases, flux decreases for some Ni, so a voltage (back emf) must aid the voltage source, to draw more current, such that the flux can be maintained.

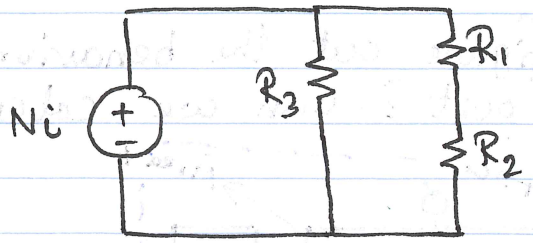
Demo: Relay.

Example:



plunger is allowed to move laterally.

Let us fix direction of 'x'



$$R_3 = \frac{g}{\mu_0 x D} \quad ; \quad R_1 = R_2 = \frac{(g-h)/2}{\mu_0 (w-x) D}.$$

$$\therefore \quad L(x) = \frac{N^2}{R_{tot}(x)} = N^2 \left[\frac{\mu_0 x D}{g} + \frac{\mu_0 (w-x) D}{g-h} \right].$$

$$= \mu_0 N^2 D \left[\frac{x}{g} + \frac{w-x}{g-h} \right]$$

∴ Looking from electrical port :

$$v(t) = L(x) \frac{di}{dt} + i \cdot \frac{\partial L(x)}{\partial x} \cdot \frac{dx}{dt}$$

$$= L(x) \frac{di}{dt} + i \left[\frac{\mu_0 N^2 D}{g} - \frac{\mu_0 N^2 D}{g-h} \right] \cdot \frac{dx}{dt}$$

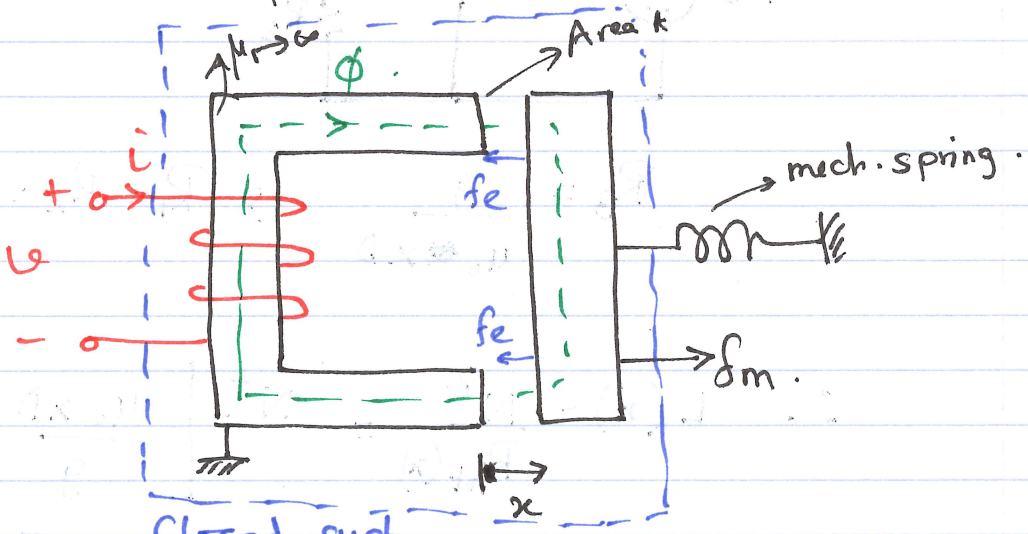
$$= L(x) \frac{di}{dt} + i \mu_0 N^2 D \left[\frac{-h}{g(g-h)} \right] \frac{dx}{dt}.$$

$0 < x < w$

$$= L(x) \frac{di}{dt} - \frac{\mu_0 N^2 D i h}{g(g-h)} \cdot \frac{dx}{dt}.$$

Bottomline : Calculate inductance, understand, the position of least stored energy.

Q. Can we figure out the behavior of the mechanical port? Can we calculate the force?



If we keep the plunger, stationary, we can inject energy through the electrical port,

$$\text{Energy, } W_m = \int v\, i\, dt = \int i\, d\lambda$$

But, we can also inject ~~energy~~ energy, by keeping the excitation, and moving the plunger to the right (positive x), by a mechanical force, f_m.

$$W_m = f_m \cdot x$$

$\therefore$ Rate of change of stored energy.

$$\frac{dW_m}{dt} = v\, i + f_m \cdot \frac{dx}{dt}$$

$\therefore$

$$\boxed{dW_m = i\, d\lambda + f_m\, dx}$$

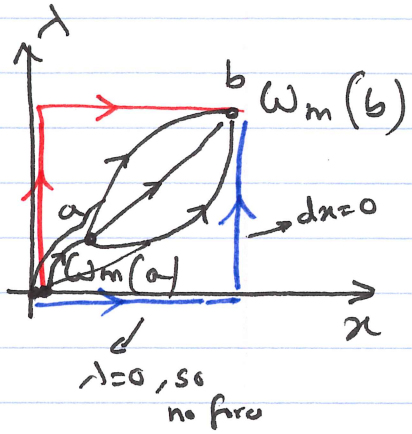
As, electrical force acts in the opposite direction of mechanical ". $f_e = -f_m$.

$$\therefore \boxed{d\omega_m = i\,d\lambda - f_e\,dx}$$

So, in $\lambda - x$ plane,

Since, the magnetic field system is lossless, it is conservative.

$\Rightarrow$ change in energy required to take the system from a to b is independent of the path. but only dependent on λ and x.



$$\omega_m(b) - \omega_m(a) = \int_a^b i\,d\lambda - f_e\,dx.$$

As, $\omega_m = f(x, \lambda)$

$$\therefore d\omega_m = \left.\frac{\partial \omega_m}{\partial x}\right|_{\lambda=\text{const}} \cdot dx + \left.\frac{\partial \omega_m}{\partial \lambda}\right|_{x=\text{const}} \cdot d\lambda.$$

$$\therefore \boxed{f_e = \left. -\frac{\partial \omega_m}{\partial x}\right|_{\lambda=\text{const}}.}$$

$$\boxed{\omega_m = \left.\int i\,d\lambda\right|_{x=\text{const}}}$$